

S3

Normal Distribution

#Learning Objectives

In this chapter, we will

- (a) understand the concept of a **continuous random variable**;
- (b) recognise the **normal distribution**, $N(\mu, \sigma^2)$, as a **continuous probability model** with mean μ and variance σ^2 ;
- (c) find probabilities of the form $P(X < x_1)$ or related probabilities, given the values x_1 , μ and σ ;
- (d) understand the **symmetry and key properties of the normal distribution curve**;
- (e) **standardise** a normal random variable using the standard normal distribution;
- (f) determine the relationship between x_1 , μ and σ given the value of $P(X < x_1)$ or a related probability;
- (g) solve problems involving the **expectation** and **variance** of a **linear transformation** of a random variable, i.e. $E(aX + bY)$ and $\text{Var}(aX + bY)$, where X and Y are independent;

3.1 Continuous Random Variables

The random variables we have learnt so far are all in discrete nature where the set of possible outcomes are countable. Unlike **discrete** random variables, there are **continuous random variables** that take values on a continuous scale. Because their outcomes cannot be listed individually, they typically represent measurements such as mass, height, distance, or time which contain an interval of real numbers.

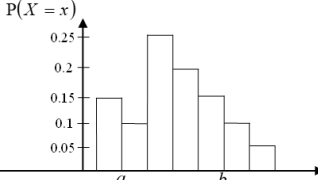
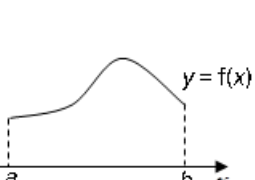
#Probability Density Function

A function that describes the likelihood for a continuous random variable to take on a given range of values is called the **probability density function (pdf)**, or **density** of the random variable (for an explanation on how this probability density function came about, please refer to Appendix A). The probability for a continuous random variable X to fall within a particular region $[a, b]$ is given by $P(a \leq X \leq b) = \int_a^b f(x) dx$, where f is the pdf of X . Since X is a continuous random variable, $P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$. It is because probability is 0 at a specific point at $x = c$. [i.e. $P(X = c) = \int_c^c f(x) dx = 0$]

The probability density function f is not just any function. It has to satisfy the following two conditions:

- (1) $f(x) \geq 0$ for all real values of x ,
- (2) $\int_{-\infty}^{\infty} f(x) dx = 1$. (Total area under the curve is 1 = Total Probability is 1)

Similar to the discrete random variable's case, we can also discuss expectation and variance of a continuous random variable. The following table gives a comparison between these two types of random variables.

Description	Notation	Discrete	Continuous
			
Probability	$P(a \leq X \leq b)$	$\sum_{x=a}^b P(X = x)$	$\int_a^b f(x) dx$
Total Prob		$\sum_{\text{all } x} P(X = x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Expectation	$\mu = E(X)$	$\sum_{\text{all } x} x P(X = x)$	$\int_{-\infty}^{\infty} x f(x) dx$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$ $\sigma^2 = \text{Var}(X)$	$\sum_{\text{all } x} x^2 P(X = x) - \mu^2$	$\int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$

#Example 1

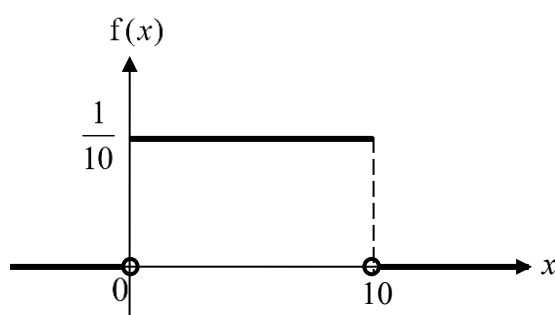
The waiting time for a shuttle bus is known to range from 0 min to 10 min. Let X denote this waiting time. Clearly, X is a continuous random variable as it may assume any values from the interval $[0, 10]$. Assume that X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{10} & 0 \leq x \leq 10, \\ 0 & \text{otherwise.} \end{cases}$$

- Sketch the graph of f .
- Find the probability that the waiting time is less than or equal to 3 minutes.
- Find $E(X)$ and $\text{Var}(X)$.

Solution

(a)



- (b) The required probability = $P(0 \leq X \leq 3) = \int_0^3 \frac{1}{10} dx = \frac{1}{10} \times 3 = \frac{3}{10}$.

- (c) By definition, $E(X) = \int_0^{10} \frac{x}{10} dx = 5$.

Alternatively, since X is **uniformly distributed** in the interval $[0, 10]$, hence $E(X)$ is equal to the midpoint of the interval. Therefore $E(X) = 5$

$$\text{Var}(X) = E(X^2) - 5^2 = \int_0^{10} \frac{x^2}{10} dx - 25 = \frac{25}{3}.$$

□

3.2 #Normal Distribution

The histogram of the IQ scores based on a large number of data are shown in Figure 1.

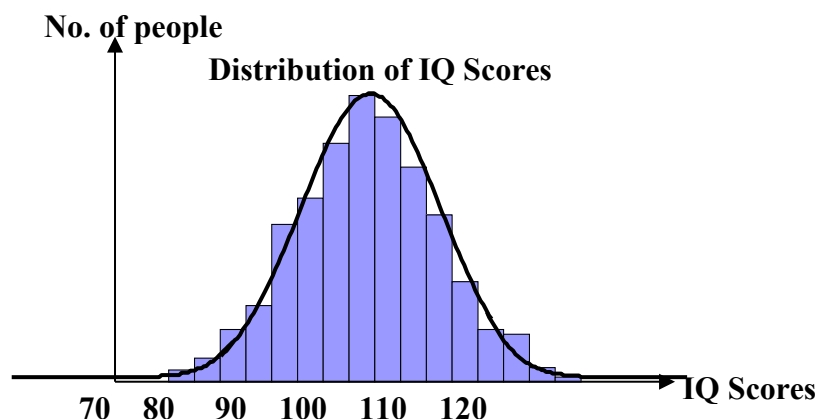


Figure 1

The histogram is almost symmetrical about a centre value and decreases from the centre. Hence, if a smooth curve is fitted onto the histogram, the curve forms a **bell-shape**. (Figure 1)

Such a curve is called the **normal distribution** curve. The application of the normal curve is evident in many industrial and educational measurements such as lifetime of a battery, mass of a chocolate bar, IQ scores and examination scores.

The random variable X has a normal distribution with mean μ and standard deviation σ , its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

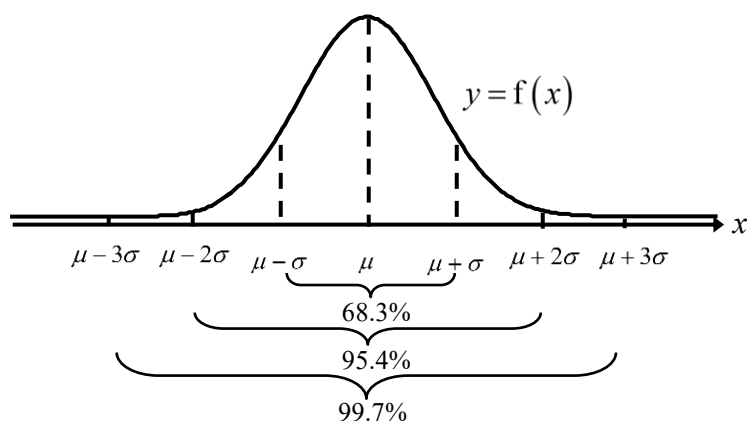
We write $X \sim N(\mu, \sigma^2)$ as σ^2 is the variance of X .

It can be proven that $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$ using the formulae for $E(X)$ and $\text{Var}(X)$ given in the table in 3.1.

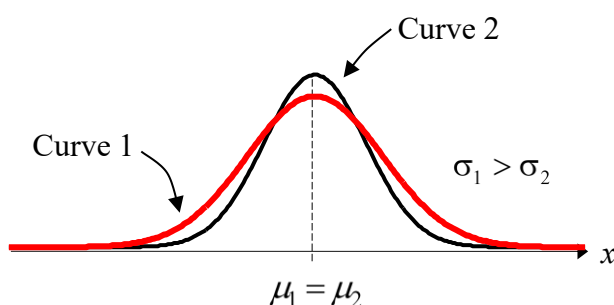
The normal distribution or specifically the "normal curve" is in bell-shape that peaks in the middle, which is the mean of the distribution. The graph also approaches the x -axis as it moves away from the peak.

For example, if x denotes the score obtained in an SAT, you may see that the extremely high and low scores are rare, while the majority of participants score somewhere close to the mean. This data can be modelled by a normal distribution.

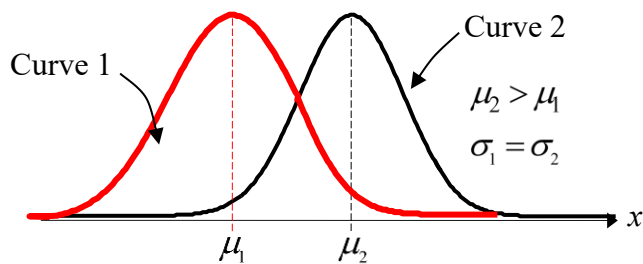
3.3 #Properties of a Normal Distribution and its Density Curve



- (i) The curve is bell-shaped, has x -axis as its horizontal asymptote and the curve extends to infinity in either direction.
- (ii) Mean = Median = Mode = μ .
- (iii) The total area under the curve is 1, i.e., $\int_{-\infty}^{\infty} f(x) dx = 1$.
- (iv) The curve is symmetrical about the line $x = \mu$.
- (v) The line of symmetry and the degree of spread is determined by the values of μ and σ respectively.
Let Curve 1 and Curve 2 represent the normal distributions of X and Y respectively, where $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$.



Same mean but different standard deviations



Different means but same standard deviation

3.4 Evaluation of Probabilities using TI84Plus CE

#Example 2

Given that $X \sim N(23, 12)$, state the expectation and the standard deviation of X . Evaluate the following probabilities:

- $P(20 \leq X \leq 25)$,
- $P(X \leq 25)$,
- $P(X > 21)$.

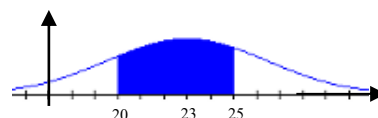
Solution

Compare $X \sim N(23, 12)$ with $X \sim N(\mu, \sigma^2) \Rightarrow E(X) = \mu = 23$ and $\sigma^2 = 12 \Rightarrow \sigma = \sqrt{12}$.

Using TI84+:

- $P(20 \leq X \leq 25) = 0.525$ (3 s.f.)

- Go to the home screen. Press **2nd** [DISTR], select **2:normalcdf**(, press **ENTER**.



- Key in 20 as the lower limit, 25 as the upper limit, 23 as the mean and $\sqrt{12}$ as the standard deviation as shown.

```
DISTR DRAW
1:normalpdf(
2:normalcdf(
3:invNorm(
4:invT(
5:tpdf(
6:tcdf(
7:x²pdf(
8:x²cdf(
9:Fpdf(
```

- Press **▾** to move the cursor to **Paste** and press **ENTER** to paste the command onto the home screen. Press **ENTER** to evaluate the answer.

```
normalcdf
lower: 20
upper: 25
μ: 23
σ: √(12)
Paste
```

```
normalcdf(20,25,23,√12)
.....0.5249105442
```

- $P(X \leq 25) = 0.718$ (3 s.f.)

- Go to the home screen. Press **2nd** [DISTR], select **2:normalcdf**(, press **ENTER**. Key in the lower limit, upper limit, the mean and the standard deviation. The lower limit is negative infinity in this case. i.e., key in **[-]** **2nd**[EE] **99**.

```
normalcdf
lower: -E99
upper: 25
μ: 23
σ: √(12)
Paste
```

Standard deviation

- ② Press $\square$ to move the cursor to **Paste** and press $\square$ to paste the command onto the home screen. And press $\square$ again to evaluate the answer.

```
normalcdf(-E99,25,23,√12)
.....
0.7181486136
```

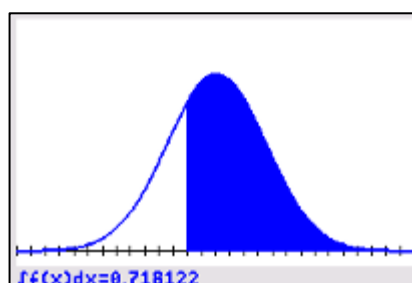
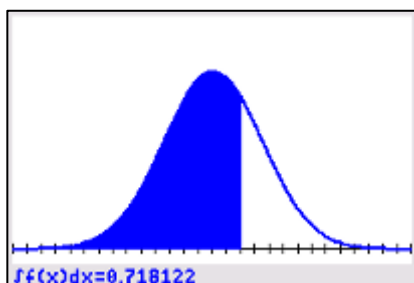
- (iii) $P(X > 21) = 0.718$ (3 s.f.)

The upper limit in this case is positive infinity.
i.e., key in $\square$ [E] 99 as the upper limit.

```
normalcdf(21,E99,23,√12)
.....
0.7181486136
```

Note

The answers in parts (ii) and (iii) are the same due to the symmetric property of the normal distribution. We notice that the values 21 and 25 are equidistant away from the mean, $\mu = 23$. (see below for illustration).



#Example 3

The mass (in grams) of a small loaf of bread, X is normally distributed with mean μ and standard deviation 20. Given that $P(X < 475) = P(X > 525)$, find $P(X < 475)$.

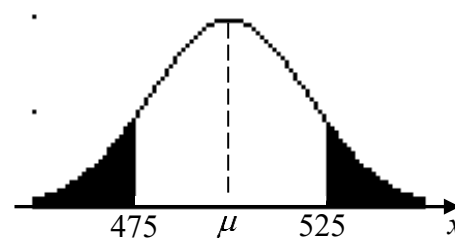
Solution

Write $X \sim N(\mu, 20^2)$.

In order to achieve $P(X < 475) = P(X > 525)$, μ must be the midpoint of 475 and 525.

By symmetry, $\mu = \frac{475 + 525}{2} = 500$

Since $X \sim N(500, 20^2)$, $P(X < 475) = 0.106$ (3 s.f.).



#Example 4

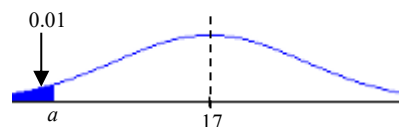
Given $X \sim N(17, 11)$, find the values of a , b and c such that

- (i) $P(X < a) = 0.01$,
- (ii) $P(X > b) = 0.3$,
- (iii) $P(c < X < d) = 0.375$ where $\frac{a+b}{2} = 17$.
- (iv) $P(15 < X < e) = 0.6$

Solution

- (i) Given $P(X < a) = 0.01$, from GC, $a = 9.28$ (3 s.f.).

- ① Go to the home screen. Press **2nd** [DISTR], select **3:invNorm**(, press **ENTER**.



- ② Key in 0.01 as the area (which is the probability required), 17 as the mean, $\sqrt{11}$ as the standard deviation and Tail: LEFT.

```
DISTR DRAW
1:normalpdf(
2:normalcdf(
3:invNorm(
4:invT(
5:tpdf(
```

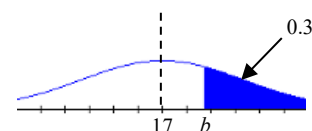
```
invNorm
area:0.01
μ:17
σ:√(11)
Tail: LEFT CENTER RIGHT
Paste
```

- ③ Press **▢** to move the cursor to **Paste** and press **ENTER** to paste the command onto the home screen. And press **ENTER** again to obtain the answer for a .

```
invNorm(0.01,17,√11,LEFT)
.....9.28437696
```

- (ii) Same as (i) but **change the area to 0.3 and Tail: RIGHT**

$$P(X > b) = 0.3, \text{ by GC } b = 18.7 \text{ (3 s.f.)}$$



```
invNorm(0.3,17,√11,RIGHT)
.....18.73923973
```

- (iii) $\frac{c+d}{2} = 17$

$\Rightarrow a$ and b are equidistant from the mean 17

$$P(c < X < d) = 0.375$$

$$P(X < c) + P(X > d) = 0.625$$

$$2P(X < c) = 0.625 \text{ due to symmetry}$$

$$P(X < c) = 0.3125 \Rightarrow c = 15.4 \text{ (3 s.f.) and hence}$$

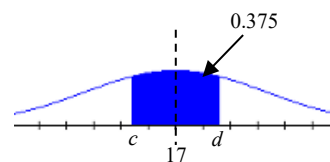
$$d = 18.6 \text{ (3 s.f.)}$$

```
invNorm(0.3125,17,√11)
.....15.37891204
```

Note

(iii) $P(c < X < d) = 0.375$

Key in 0.375 as the area (which is the probability required), 17 as the mean and $\sqrt{11}$ as the standard deviation. For Tail, choose "CENTER". [See (*) below]



Press $\square$ to move the cursor to **Paste** and press **ENTER** to paste the command onto the home screen. And press **ENTER** again to obtain the answer for a and b .

```
invNorm
area:0.375
μ:17
σ:√(11)
Tail: LEFT CENTER RIGHT
Paste
```

Important: "TAIL: CENTER" can only be used when a and b are symmetrical about the mean.

```
invNorm(0.375,17,√11,CENTR
{15.37891204 18.62108796}..
```

(iv) $P(15 < X < e) = 0.6$

Do not use "Tail: CENTER" under invNorm because the range is not symmetrical about 17. We should get $P(X < 15) \approx 0.27325$.

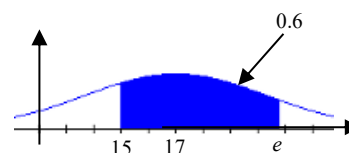
Therefore, we should find c in such a way that

$$P(X < e) = P(X < 15) + P(15 < X < e)$$

$$= 0.6 + 0.27325$$

$$= 0.87325$$

Using invNorm, by GC $e = 20.8$ (3 s.f.)



TI84 Plus CE allows the computation of inverse cumulative normal distribution for a given area under the normal distribution curve specified by μ and σ where the optional argument tail could be **LEFT** $(-\infty, x)$, **CENTER** $[-x, x]$ or **RIGHT** (x, ∞) for $x \in \mathbb{R}$

The **TI84+** allows for the **LEFT** $(-\infty, x)$ only.

Example 5

The marks obtained in a mathematics examination are found to be normally distributed with mean 60 and standard deviation 12.

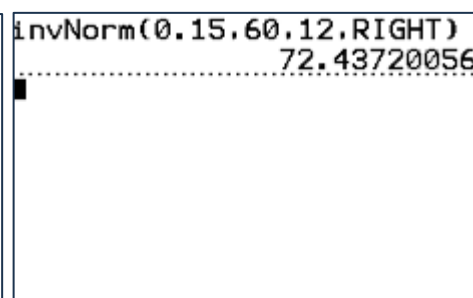
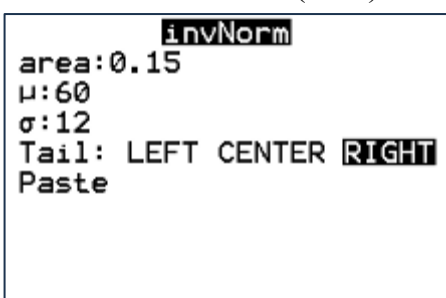
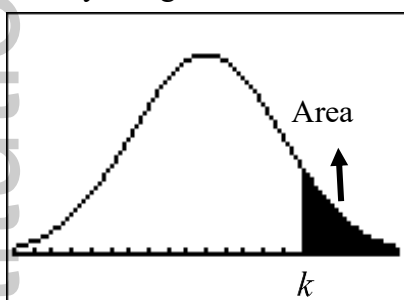
- Find the probability that when the mark obtained is doubled, it is higher than 70.
- Find the probability that the mark obtained is between 45 and 60.
- Find the mark for grade A if 15% of the candidates are to obtain grade A.
- Find, to the nearest whole number, the least mark for grade A if at most 15% of the candidates are to obtain grade A.

Solution

Let X be the mark in a mathematics examination. Then $X \sim N(60, 12^2)$.

- $P(2X \geq 70) = P(X \geq 35) = 0.981$ (3 s.f.)
- $P(45 < X < 60) = 0.394$ (3 s.f.)
- Let k be least mark for grade A, then $P(X \geq k) = 0.15$.

By using the Normal Distribution curve, $k = 72.4$ (3 s.f.).



- $P(X \geq k) \leq 0.15$

By using **invNorm** in GC, $k = 72.4$.

Therefore, least value of k is 73.

$k > 72.5$ (3 s.f.)

Alternative Method for (d): Using the table
(only when k is an integer)

- Press **[Y=]**. In Y_1 , press **[2nd][VARS]** and select **2:normalcdf**.
- Key in **[] [EE] 99** as lower limit, press **[X,T,θ,n]** to enter ' X ' as upper limit, 60 as mean and 15 as standard deviation.
- Press **[]** to move the cursor to **Paste** and press **[ENTER]** to paste the command into Y_1 as shown.
- Press **[2nd][TABLE]** and scroll down the table for the value of X that gives Y_1 more than or equal to 0.85.

From table, $P(X < 72) = 0.8413$

$$P(X < 73) = 0.8607$$

$$P(X < 74) = 0.8783$$

Therefore least $k = 73$.

Plot1	Plot2	Plot3
$Y_1 = \text{normalcdf}(-1E99, X, 60, 15)$		
$Y_2 =$		
$Y_3 =$		
$Y_4 =$		
$Y_5 =$		
$Y_6 =$		
$Y_7 =$		
$Y_8 =$		
$Y_9 =$		

X	Y1			
70	0.7977			
71	0.8203			
72	0.8413			
73	0.8607			
74	0.8783			
75	0.8944			
76	0.9088			
77	0.9217			
78	0.9332			
79	0.9433			
80	0.9522			

X=73

3.5 #Standard Normal Distribution

If a random variable which is normally distributed has mean 0 and variance 1, it is called the **standard normal random variable**. Its distribution is called the **standard normal distribution**. We denote the standard normal random variable by Z and write $Z \sim N(0, 1)$.

Note

We use Z to denote the standard normal random variable instead of the usual notation X .

Example 6

Evaluate

- (i) $P(-1 < Z < 2)$,
- (ii) $P(Z < -2)$,
- (iii) $P(Z > 2)$,
- (iv) $P(|Z| > 1)$.

Solution

(i) $P(-1 < Z < 2) = 0.819$ (3 s.f.).

- ① Press **2nd** [DISTR], select **2:normalcdf** , press **ENTER**.
- ② Key in the lower limit as -1 and the upper limit as 2 and leave the mean as 0 and the standard deviation as 1 .
- ③ Press **▼** to move the cursor to **Paste** and press **ENTER** to paste the command onto the home screen. Press **ENTER** again to evaluate the answer.

```
normalcdf
lower: -1
upper: 2
μ: 0
σ: 1
Paste
```

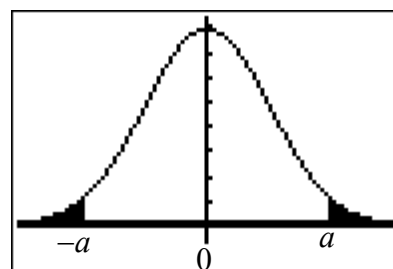
```
normalcdf(-1.2,0.1)
.....0.8185946784
```

(ii) $P(Z < -2) = 0.0228$ (3 s.f.)

(iii) $P(Z > 2) = 0.0228$ (3 s.f.)

The answers in parts (ii) and (iii) are equal because -2 and 2 are symmetrical about the mean 0 .

In fact, $P(Z < -a) = P(Z > a)$ for all real values of a where $a \geq 0$.



$$\begin{aligned}
 \text{(iv)} \quad & P(|Z| > 1) \\
 &= P(Z < -1) + P(Z > 1) \\
 &= 2P(Z < -1) = 0.317
 \end{aligned}$$

```
normalcdf(-E99, -1, 0, 1)*2
0.3173105191
```

Alternatively,

$$P(|Z| > 1) = P(Z < -1) + P(Z > 1) = 2P(Z > 1) = 0.317 \text{ (3 s.f.)}$$

□

#Example 7

Find the value of a where $P(1 < Z < a) = 0.1$.

Solution

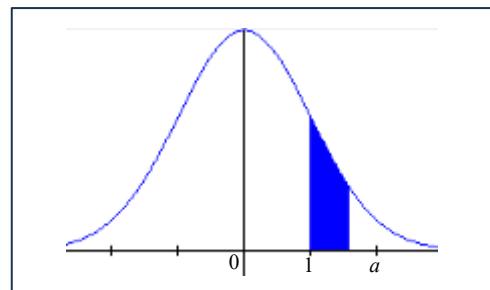
First, we rewrite $P(1 < Z < a) = P(Z < a) - P(Z \leq 1)$.

From GC, $P(Z \leq 1) = 0.84134$.

Thus $P(Z < a) - 0.84134 = 0.1 \Rightarrow P(Z < a) = 0.94134$.

From GC, $a = 1.57$ (3 s.f.)

- ① Go to home screen and press **2nd** [DISTR], select **3:invNorm** , press **ENTER**.
- ② Key in 0.9 as the area, choose “LEFT” in Tail and press **▾** to move the cursor to **Paste**. Press **ENTER** to paste the command onto the home screen.
- ③ Press **ENTER** to obtain the answer for a .



```
invNorm
area:0.94134
μ:0
σ:1
Tail: LEFT CENTER RIGHT
Paste
```

```
invNorm(0.94134, 0, 1, LEFT)
1.566122249
```

3.6 Relationship between $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0, 1)$

If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$.

This substitution is called **standardisation** and we say that X is standardised to the standard normal variable Z .

To find the probability involving X , we can always standardize X and find the equivalent probability involving Z . This is as shown below:

$$\begin{aligned} & P(a < X < b) \\ &= P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right) \quad (\text{subtract } \mu \text{ and then divide by } \sigma \text{ for each term}) \\ &= P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) \quad (\text{replace } \frac{X - \mu}{\sigma} \text{ by } Z) \end{aligned}$$

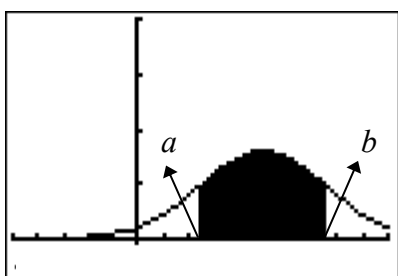


Figure 2: $X \sim N(\mu, \sigma^2)$

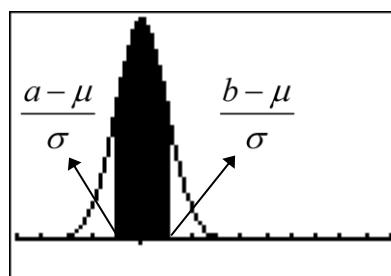


Figure 3: $Z \sim N(0, 1)$

The area under the p.d.f. of X from $x = a$ to $x = b$ is the same as the area under p.d.f. of Z from $z = \frac{a - \mu}{\sigma}$ to $z = \frac{b - \mu}{\sigma}$ as shown in Figures 2 and 3.

Note

The use of standardisation is to tackle problems involving normal distributions with unknown mean and variance. This is shown in the following examples.

Example 8 (N2008/P1/Q7)

An examination is marked out of 100. It is taken by a large number of candidates. The mean mark, for all candidates, is 72.1 and the standard deviation is 15.2. Give a reason why a normal distribution, with this mean and standard deviation, would not give a good approximation to the distribution of marks.

Solution

It is not a good approximation as $P(X > 100) = 0.0332$ (3 s.f.) which is not small if the candidature is large. For example, in a school with about 1000 students in a cohort, $0.0332 \times 1000 \approx 33$, which means there are 33 students (more than one class) with marks more than 100.

#Example 9

Given $X \sim N(\mu, \sigma^2)$, find $P(X \leq \mu + 2\sigma)$.

Solution

Even though mean and standard deviation are unknown, we can still find the answer by standardising X .

$$P(X \leq \mu + 2\sigma) = P\left(\frac{X - \mu}{\sigma} \leq \frac{\mu + 2\sigma - \mu}{\sigma}\right) = P(Z \leq 2) = 0.977 \text{ (3 s.f.)}$$

□

Example 10

The time, in minutes, that 16 year old students spend on their mathematics tutorial in a day follows a normal distribution with mean μ and variance σ^2 . It is known that 5% of the students spend less than 40 minutes and 60% of them spend more than an hour on their mathematics tutorial in a day. Find the values of μ and σ , correct to 2 decimal places.

Solution

Let X be the time (in minutes) that 16 year old students spend on their mathematics tutorial in a day. Then $X \sim N(\mu, \sigma^2)$.

It is given that $P(X < 40) = 0.05$.

$$P\left(\frac{X - \mu}{\sigma} < \frac{40 - \mu}{\sigma}\right) = 0.05$$

$$P\left(Z < \frac{40 - \mu}{\sigma}\right) = 0.05$$

$$\frac{40 - \mu}{\sigma} = -1.644853626 \text{ --- (1)}$$

It is given that $P(X > 60) = 0.6$.

$$P\left(Z > \frac{60 - \mu}{\sigma}\right) = 0.6$$

$$1 - P\left(Z \leq \frac{60 - \mu}{\sigma}\right) = 0.6$$

$$P\left(Z \leq \frac{60 - \mu}{\sigma}\right) = 0.4$$

$$\frac{60 - \mu}{\sigma} = -0.2533471011 \text{ --- (2)}$$

With **TI84 Plus CE**, we can directly find

$$P\left(Z > \frac{60 - \mu}{\sigma}\right) = 0.6$$

```

invNorm
area:0.6
μ:0
σ:1
Tail: LEFT CENTER RIGHT
invNorm(0.6,0,1,RIGHT)
.....-0.2533471011
    
```

Recall that we need to change to the $P(Z \leq a)$ form before we can use the invNorm function for TI84+

Solving equations (1) and (2) give $\sigma = 14.3729 \approx 14.37$, $\mu = 63.641 \approx 63.6$. (3 s.f.)

□

3.7 #Linear Combinations of Independent Random Variables which are Normally Distributed

An important property of **Normal distributions** is that any linear combination of independent normal random variables is itself normally distributed.

The expectation and variance of this new variable are calculated using the formulas previously introduced in Chapters 2 and 3.1:

- (i) $E(aX + b) = a E(X) + b$,
- (ii) $E(aX \pm bY) = a E(X) \pm b E(Y)$,
- (iii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$,
- (iv) $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$, if X and Y are also independent.

These results are used to derive the sum and difference of normal random variables.

Sum and Difference of Independent Normal Random Variables

- (a) If X and Y are independent random variables such that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, then $aX \pm bY$ are also normally distributed such that

$$aX \pm bY \sim N(a\mu_1 \pm b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$$

where a and b are non-zero constants.

- (b) If $X_1, X_2, \dots, X_n$ are n independent normal random variables such that $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, \dots, n$, then $X_1 + X_2 + \dots + X_n$ is also normally distributed such that
- $$X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2).$$

Proof

The proof that $X \pm Y$ and $X_1 + X_2 + \dots + X_n$ are normally distributed is beyond our syllabus. However, their expectations and variances can be derived from the results given earlier on:

- (a) $E(aX \pm bY) = aE(X) \pm bE(Y) = a\mu_1 \pm b\mu_2$
 $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) = a^2\sigma_1^2 + b^2\sigma_2^2$

X and Y are independent.

- (b) $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n) = \mu + \mu + \dots + \mu = n\mu$
 $\text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)$
 $= \sigma^2 + \sigma^2 + \dots + \sigma^2 = n\sigma^2$

$X_1, X_2, \dots, X_n$ are independent.

□

#Example 11

For a particular subject, the examination marks of the boys are normally distributed with a mean of 55 and standard deviation 11. The marks of the girls are also normally distributed with a mean of 58 and standard deviation 8. If one boy and one girl are chosen at random from the complete list of candidates, calculate the probability that

- the sum of the two marks is at most 100,
- the girl's mark exceeds the boy's mark,
- the difference between the two marks is more than 20,

Solution

Let X be the examination mark of a boy and Y be the examination mark of a girl.

Then $X \sim N(55, 11^2)$ and $Y \sim N(58, 8^2)$.

$$(i) \quad E(X + Y) = E(X) + E(Y) = 55 + 58 = 113$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 11^2 + 8^2 = 185$$

Then $X + Y \sim N(113, 185)$.

$$P(X + Y \leq 100) = 0.170 \quad (3 \text{ s.f.})$$

$$(ii) \quad P(Y > X) = P(Y - X > 0)$$

$$E(Y - X) = E(Y) - E(X) = 3 \text{ and } \text{Var}(Y - X) = \text{Var}(Y) + (-1)^2 \text{Var}(X) = 185$$

Then $Y - X \sim N(3, 185)$.

$$P(Y - X > 0) = 0.587 \quad (3 \text{ s.f.})$$

$$(iii) \quad \text{The difference between the two marks is expressed as } |Y - X|.$$

$$P(|Y - X| > 20) = P(Y - X > 20) + P(Y - X < -20) = 0.151 \quad (3 \text{ s.f.})$$

□

#Example 12

The mass of tea in a peppermint teabag has a normal distribution with mean 4.1 g and standard deviation 0.12 g. Find the probability that five randomly chosen peppermint teabags contain a total of more than 21 g of tea.

Solution

Let X be the mass of tea in a peppermint teabag. Then $X \sim N(4.1, 0.12^2)$.

Also $X_i \sim N(4.1, 0.12^2)$ for $i = 1, 2, 3, 4, 5$.

Total mass of five randomly chosen peppermint teabags = $X_1 + X_2 + X_3 + X_4 + X_5$.

$$E(X_1 + X_2 + \dots + X_5) = E(X_1) + E(X_2) + \dots + E(X_5) = 5(4.1) = 20.5 \quad \boxed{5\mu}$$

$$\text{Var}(X_1 + X_2 + \dots + X_5) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_5) = 5(0.12^2) = 0.072$$

Then $X_1 + X_2 + X_3 + X_4 + X_5 \sim N(20.5, 0.072)$.

$$\text{Required probability} = P(X_1 + X_2 + X_3 + X_4 + X_5 > 21) = 0.0312 \quad (3 \text{ s.f.}) \quad \boxed{5\sigma^2}$$

□

Note

The total mass of five randomly chosen peppermint teabags **must not** be mistaken as $5X$. The latter represents five times the mass of a peppermint teabag while the former represents the mass of 5 possibly different-mass teabags and is written as $X_1 + X_2 + X_3 + X_4 + X_5$.

#Example 13

In a farm, the mass C of a chick is normally distributed with mean 99 g and standard deviation 14 g. The mass D of a duckling is normally distributed with mean 113 g and standard deviation 16 g. C and D are independent of each other.

- Find the probability that twice the mass of a chick will be more than the mass of a duckling by at least 20 g.
- If the probability of the total mass of two randomly chosen chicks exceeding k g is 0.9, find value of k .
- Find the probability that average mass of two randomly chosen chicks and three randomly chosen ducklings is less than 100 g.

Solution

$C \sim N(99, 14^2)$ and $D \sim N(113, 16^2)$

- We want to find $P(2C - D \geq 20)$.

$$E(2C - D) = 2E(C) - E(D) = 2(99) - 113 = 85$$

$$\text{Var}(2C - D) = 2^2 \text{Var}(C) + (-1)^2 \text{Var}(D) = 4(14^2) + 16^2 = 1040$$

Then $2C - D \sim N(85, 1040)$.

From GC, $P(2C - D \geq 20) = 0.978$ (3 s.f.)

$$\text{Var}(2C) = 2^2 \text{Var}(C)$$

- Total mass of two randomly chosen chicks = $C_1 + C_2$.

Then $C_1 + C_2 \sim N(2 \times 99, 2 \times 14^2) \Rightarrow C_1 + C_2 \sim N(198, 392)$

$$\Rightarrow P(C_1 + C_2 > k) = 0.9$$

$$\Rightarrow P(C_1 + C_2 \leq k) = 0.1$$

From GC, $k = 173$.

In TI84 Plus CE, under invNorm, directly key in 0.9 as the area, mean as 2×99 and standard deviation as $\sqrt{2 \times 14^2}$, choose Tail as "RIGHT" to find k .

- We want to find $P\left(\frac{C_1 + C_2 + D_1 + D_2 + D_3}{5} < 100\right) = P(C_1 + C_2 + D_1 + D_2 + D_3 < 500)$.

$$E(C_1 + C_2 + D_1 + D_2 + D_3) = 2E(C) + 3E(D) = 2(99) + 3(113) = 537$$

$$\text{Var}(C_1 + C_2 + D_1 + D_2 + D_3) = 2\text{Var}(C) + 3\text{Var}(D) = 2(14^2) + 3(16^2) = 1160$$

Let $T = C_1 + C_2 + D_1 + D_2 + D_3$. Then $T \sim N(537, 1160)$.

Thus $P(T < 500) = 0.139$ (3 s.f.)

□

Note

In general, for the normal distribution to be an appropriate model for a random variable X which represents a scalar quantity (such as mass, length, distance etc.), $P(X < 0)$ is necessarily negligible and we conclude that it is not required to consider $P(0 \leq X < k)$ when we are evaluating $P(X < k)$.

In Example 13, $P(0 \leq T < 500) = 0.1386604855$ and $P(T < 500) = 0.1386604855$ which shows both probabilities are the same for at least the first 10 decimal places.

#Example 14

The mass (in kg) of a salmon fish is a random variable with the distribution $N(2, 0.25)$. The mass (in kg) of a tuna fish is a random variable with the distribution $N(3, 0.64)$.

- Find the probability that the mass of 4 salmon fish is more than 3 times the mass of a tuna fish.
- Find the probability that the mass of 4 salmon fishes is more than that of 3 randomly chosen of tuna fishes.
- State an assumption needed for your calculation for parts (i) and (ii) to be valid.
- Given that 75% of the time, the total mass of n randomly chosen tuna fish is less than 50 kg, find the value of n .

Solution

Let S be the mass of a salmon fish and T be the mass of a tuna fish.

Then $S \sim N(2, 0.25)$ and $T \sim N(3, 0.64)$.

- $$E(S_1 + S_2 + S_3 + S_4 - 3T) = 4(2) - 3(3) = -1 \text{ and}$$

$$\text{Var}(S_1 + S_2 + S_3 + S_4 - 3T) = 4(0.25) + (-3)^2(0.64) = 6.76$$

Then $S_1 + S_2 + S_3 + S_4 - 3T \sim N(-1, 6.76)$.

$P(S_1 + S_2 + S_3 + S_4 - 3T > 0) = 0.350$ (3 s.f.)

- $$E(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3)) = 4(2) - 3(3) = -1 \text{ and}$$

$$\begin{aligned} \text{Var}(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3)) \\ = \text{Var}(S_1 + S_2 + S_3 + S_4) + (-1)^2 \text{Var}(T_1 + T_2 + T_3) = 4(0.25) + 3(0.64) = 2.92 \end{aligned}$$

Caution: $T_1 + T_2 + T_3 \neq 3T$

Then $S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3) \sim N(-1, 2.92)$.

$P(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3) > 0) = 0.279$ (3 s.f.).

- The assumption is that S and T are independent of each other.
- We want to find n such that $P(T_1 + T_2 + \dots + T_n < 50) = 0.75$.

Let X be $T_1 + T_2 + \dots + T_n$.

$$E(X) = 3n \text{ and } \text{Var}(X) = 0.64n$$

Then $X \sim N(3n, 0.64n)$.

$$P(X < 50) = 0.75$$

As the mean and variance in the distribution of X are unknown, we standardise X .

$$P\left(Z < \frac{50 - 3n}{\sqrt{0.64n}}\right) = 0.75$$

$$\frac{50 - 3n}{\sqrt{0.64n}} = 0.6744897495$$

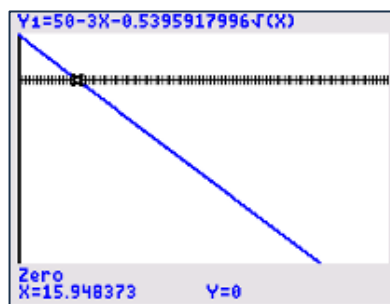
$$50 - 3n = 0.6744897495\sqrt{0.64n}$$

$$50 - 3n - 0.5395917996\sqrt{n} = 0$$

invNorm(0.75, 0, 1)

We can solve the equation by plotting the graph using GC.

From GC, $n = 15.948 \Rightarrow n = 16$ (for integer answer).



Alternatively, we can solve for n given $X \sim N(3n, 0.64n)$ and $P(X < 50) = 0.75$ by using the equation editor and table. Thus from GC,

n	$P(X < 50)$
15	0.94671
16	0.73401
17	0.38088

We conclude that $n = 16$.

- ① Press $\boxed{Y=}$ $\boxed{2nd}$ $\boxed{[DISTR]}$ and select **2:normalcdf**, press $\boxed{ENTER}$,
- ② Key in $\boxed{-}$ $\boxed{EE}$ 99 as lower limit, 50 as upper limit, 3X as mean and $\sqrt{(0.64X)}$ as standard deviation.
- ③ Press $\boxed{\downarrow}$ to move the cursor to **Paste** and press $\boxed{ENTER}$ to paste the command onto the equation editor.

```
normalcdf
lower: -1E99
upper: 50
μ: 3X
σ: sqrt(0.64X)
Paste
```

- ④ Key in 0.75 as Y_2 .
- ⑤ Press $\boxed{2\text{nd}} \boxed{\text{TABLE}}$ and scroll down the table for the value of X that gives Y_1 closest to 0.75.

Plot1	Plot2	Plot3
$\text{Y}_1 = \text{normalcdf}(-1\text{E}99, 50.3, \text{X})$		
$\text{Y}_2 = 0.75$		
$\text{Y}_3 =$		
$\text{Y}_4 =$		
$\text{Y}_5 =$		
$\text{Y}_6 =$		
$\text{Y}_7 =$		
$\text{Y}_8 =$		

X	Y ₁	Y ₂			
12	1	0.75			
13	0.9999	0.75			
14	0.9962	0.75			
15	0.9467	0.75			
16	0.734	0.75			
17	0.3809	0.75			
18	0.1193	0.75			
19	0.0224	0.75			
20	0.0026	0.75			
21	2E-4	0.75			
22	1E-5	0.75			

X=16

Summary

- A random variable X which is normally distributed with mean μ and variance σ^2 , is denoted by $X \sim N(\mu, \sigma^2)$.
- The distribution is bell-shaped and symmetrical about the vertical line $x = \mu$.
- mean = median = mode = μ
- If a normal distribution $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$ is the standard normal distribution. Then, $P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$.
- If $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent random variables, then $aX \pm bY \sim N(a\mu_1 \pm b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$ where a and b are constants.
- If $X_1, X_2, \dots, X_n$ are independent random variables such that $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, 3, \dots, n$, then $X_1 + X_2 + \dots + X_n$ is also normally distributed such that $X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2)$.

Appendix A Probability Density Function

The masses of 200 six-month-old babies are collected. To visualise the distribution of the mass of a six-month old baby (which is a continuous random variable), the data collected are grouped into 5 classes with a class width of 1 kg each. The frequency table and the histogram corresponding to the data (Figure 1) are shown below.

x , Mass (kg)	$5 \leq x < 6$	$6 \leq x < 7$	$7 \leq x < 8$	$8 \leq x < 9$	$9 \leq x \leq 10$
Frequency, f	20	48	80	36	16

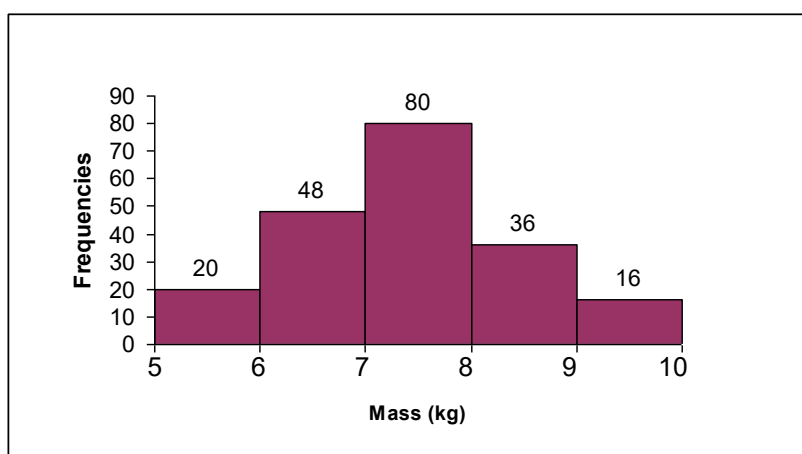


Figure 1

The relative frequency table and histogram of the data is shown in Figure 2 below.

x , Mass (kg)	$5 \leq x < 6$	$6 \leq x < 7$	$7 \leq x < 8$	$8 \leq x < 9$	$9 \leq x \leq 10$
Relative frequency, $f / 200$	0.1	0.24	0.4	0.18	0.08

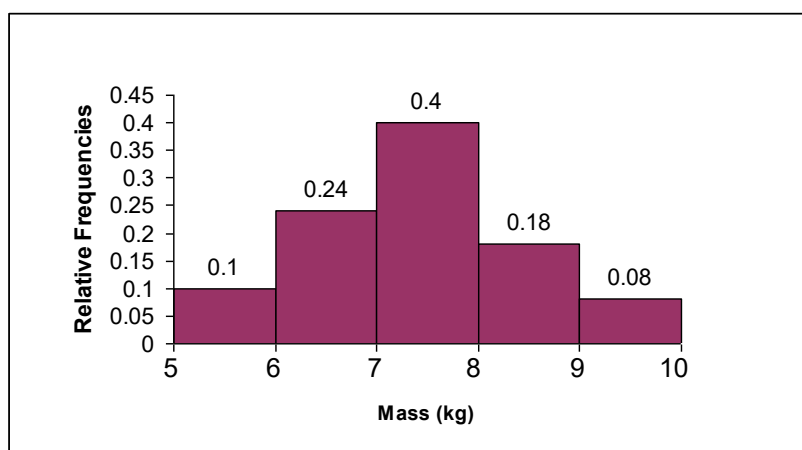


Figure 2

In a relative frequency histogram,

- the heights of the rectangles are the relative frequencies,
- the area of each rectangle is the probability of the variable lying within the respective class interval,
- the total area of the rectangles is equal to 1.

A relative frequency histogram is preferred to a histogram (which only shows absolute frequencies rather than relative ones) because the area of each rectangle gives the probability of the mass of a randomly selected six-month-old baby lying within that respective class interval. To give a better representation of the distribution of the masses, the data can be further grouped into more classes. The relative frequency histogram of the masses when $k = 20$ is shown in Figure 3 below.

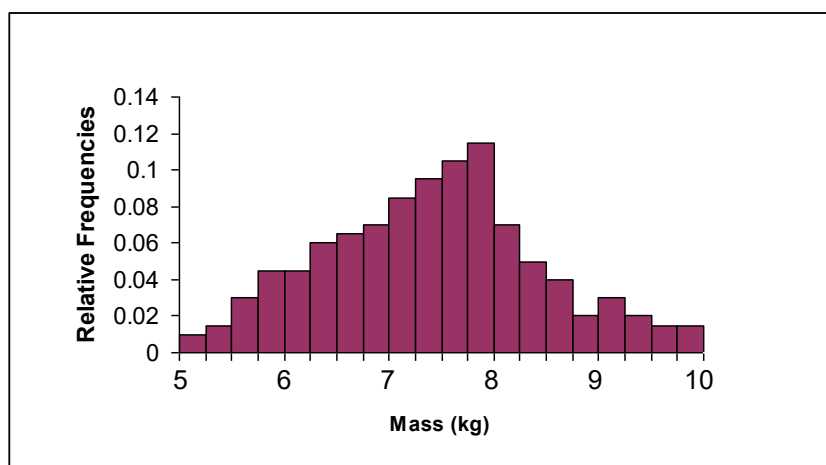


Figure 3

A relative frequency of k classes will have a class width of $\frac{5}{k}$ kg each where k is an integer.

Let us now consider what happens to the relative frequency histogram when k tends to infinity and the class width approaches zero. As k increases, the histogram will become closer to a graph of some function $f(x)$ (whose graph is a continuous curve) such that the area under this curve equals to one. This is because the total area of the rectangles in the relative frequency histogram for any given value of k is always equal to the constant 1. So, it is a reasonable consequence that the area under the curve $y = f(x)$, being the limit of a sequence of these relative frequency histograms, is equal to 1. The function $f(x)$ which describes the distribution of the mass of a six-month-old baby is called a **probability density function (p.d.f.)**.

Statistics Tutorial 3

1. N2017/P2/Q10

A small component for a machine is made from two metal spheres joined by a short metal bar. The masses in grams of the spheres have the distribution $N(20, 0.5^2)$.

- (i) Find the probability that the mass of a randomly selected sphere is more than 20.2 grams. [1]

In order to protect them from rusting, the spheres are given a coating which increases the mass of each sphere by 10%.

- (ii) Find the probability that the mass of a coated sphere is between 21.5 and 22.45 grams. State the distribution you use and its parameters. [3]

- (iii) The masses of the metal bars are normally distributed such that 60% of them have a mass greater than 12.2 grams and 25% of them have a mass less than 12 grams. Find the mean and standard deviation of the masses of metal bars. [4]

- (iv) The probability that the total mass of a component, consisting of two randomly chosen coated spheres and one randomly chosen bar, is more than k grams is 0.75. Find k , stating the parameters of any distribution you use. [4]

2. RVHS Prelim 9758/2018/02/Q10

Gary likes to take part in duathlons where there is a total running distance of 15 km and a total cycling distance of 36 km. His timings, in minutes, for running and cycling are modelled as having independent normal distributions with means and standard deviations as shown in the table.

	Mean	Standard Deviation
Timing for running (min)	100	15
Timing for cycling (min)	μ	σ

- (i) The probability that he completes his cycling in less than an hour is equal to the probability that he completes his cycling after 2 hours. Write down the value of μ . [1]
- (ii) The probability that he completes his cycling under 80 minutes is 0.158655. Show that $\sigma = 10$. [2]

- (iii) Find the probability that thrice the time needed for him to complete cycling is more than an hour from twice the time needed for him to complete the running. [2]
- (iv) Find the probability that his timing for cycling is less than 80 minutes and his timing for running is less than 90 minutes. [2]
- (v) Show that the probability of Gary finishing the duathlon under 170 minutes is 0.134. Give a reason why this probability is greater than the probability calculated in part (iv). [2]
- Gary's target timing for completing a duathlon is under 2 hours 50 minutes. Over the past two years, he has already completed 9 duathlons.
- (vi) Find the probability that he achieves his target timing in at least 3 but fewer than 6 of the duathlons over the last two years. State an assumption needed for your calculation to be valid. [4]

3. TMJC Prelim 9758/2020/02/Q9

- (i) X and Y are independent random variables with the distributions $N(\mu, 5^2)$ and $N(12, \sigma^2)$ respectively. Given that $P(X + Y < 12) = 0.25$ and $P(Y - X > 16) = 0.085$, show that $\mu = 3.9$ correct to 1 decimal place, and calculate the value of σ . [6]
- (ii) The random variable K is the sum of n independent observations of X . State the approximate value of $P(K > 3n)$ as n becomes very large, justifying your answer. [2]

4. ACJC Prelim 9758/2017/02/Q10

- (a) An examination taken by a large number of students is marked out of a total score of 100. It is found that the mean is 73 marks and that the standard deviation is 15 marks.
Give a reason why the normal distribution is not a good model for the distribution of marks for the examination. [1]

- (b) The interquartile range of a distribution is the difference between the upper and lower quartile values for the distribution. The lower quartile value, l , of a distribution X , is such that $P(X < l) = 0.25$. The upper quartile value, u , of the same distribution is such that $P(X < u) = 0.75$.

The marks of another examination is known to follow a normal distribution. If a student who scores 51 marks is at the 80th percentile, and the interquartile range is found to be 10.8 marks, find the mean mark and the standard deviation of the marks scored by students who took the examination. [5]

- (c) In a third examination, the marks scored by students are normally distributed with a mean of 52 marks and a standard deviation of 13 marks.

(i) If 50 is the passing mark and 289 students are expected to pass, how many candidates are there? [2]

(ii) Find the smallest integer value of m such that more than 90% of the candidates will score within m marks of the mean. [3]

5. **YIJC Prelim 9758/2022/02/Q12**

In this question you should state the parameters of any distributions that you use.

A supermarket sells two types of sugar. White sugar is sold in packets with the labelled mass of 1 kg. The mass of a packet of white sugar may be regarded as a normally distributed random variable with mean 1.05 kg and standard deviation 0.03 kg.

- (a) What mass is exceeded by 80% of the packets of white sugar? Give your answer correct to 3 decimal places. [1]
- (b) A packet of white sugar that weighs less than 98% of the labelled mass is considered underweight. Find the probability that at most 1 out of 10 randomly chosen packets of white sugar is underweight. [3]

The masses of packets of brown sugar are normally distributed with mean m kg and standard deviation 0.05 kg and the masses of packets of white sugar and brown sugar have independent normal distributions.

- (c) Given that the probability that the total mass of 4 randomly chosen packets of white sugar exceeds twice the mass of a randomly chosen packet of brown sugar is 0.15, find m . [5]

It is now given that $m = 2.03$.

- (d) Find the probability that the average mass of 4 randomly chosen packets of white sugar and 5 randomly chosen packets of brown sugar exceeds 1.6 kg. [4]

6. **N2018/P2/Q10 (modified)**

In this question you should state the parameters of any distributions that you use.

A manufacturer produces specialist light bulbs. The masses in grams of one type of light bulb have the normal distribution $N(50, 1.5^2)$.

- (i) Sketch the distribution for masses between 40 grams and 60 grams. [2]
- (ii) Find the probability that the mass of a randomly chosen bulb is less than 50.4 grams. [1]
- (iii) Two light bulbs are randomly chosen. Find the probability that one of the light bulbs has mass less than 50.4 g and the other has mass more than 50.4 g. [3]

Each light bulb is packed into a randomly chosen box. The masses of the empty boxes have the distribution $N(75, 2^2)$.

- (iv) Find the probability that the total mass of 4 randomly chosen empty boxes is more than 297 grams. [2]
- (v) Find the probability that the total mass of a randomly chosen light bulb and a randomly chosen box is between 124.9 and 125.7 grams. [3]

In order to protect the bulbs in transit each bulb is surrounded by padding before being packed in a box. The mass of the padding is modelled as 30% of the mass of the bulb.

- (vi) The probability that the total mass of a box containing a bulb and padding is more than k grams is 0.9. Find k . [4]
- (vii) Find the probability that the total mass of 4 randomly chosen boxes, each containing a bulb and padding, is more than 565 grams. [3]

7. **DHS/2018/C2BT2/P2/Q4(modified)**

In this question, you should state clearly the values of the parameters of any normal distribution you use.

The luggage handled by DH airline can be classified as light or heavy. The masses of these luggage, in kg, are modelled as having independent normal distributions with means and standard deviations as shown in the following table.

	Mean	Standard deviation
Light	14	σ
Heavy	27	2.5

- (i) Given that 20% of the light luggage have masses exceeding 15 kg, show that $\sigma = 1.19$ correct to 3 significant figures. [2]
- (ii) Find the range of values of a such that the probability of the mass of a randomly chosen heavy luggage greater than a kg is larger than 0.9. [2]
- (iii) For health and safety reasons, workers are not supposed to carry consecutive heavy luggage exceeding 32 kg each. A worker randomly picks three heavy luggage, one after another, to carry. Find the probability that the safety rule is breached. [2]
- (iv) Find the probability that the difference of the masses of two randomly chosen light luggage is less than 0.5 kg. [2]

8. **RVHS Prelim 9758/2023/02/Q11 (modified)**

Rectangular blocks of wood undergo crafting and lacquering to become ornaments. The masses of the rectangular blocks, X g, are known to have normal distribution with mean 300 g and standard deviation 20 g. Crafting the company logo on a block reduces its mass by 10%. After that, a layer of lacquer is applied to the crafted wooden block that increases its mass by 5%.

- (i) Show that the probability that a piece of completed ornament has mass between 290 g and 350 g is approximately 0.365. [3]

The ornament is placed into a gift box before it is sold. The masses of the gift boxes modelled as 100α % of the masses of the ornaments.

- (ii) Given that the probability of a randomly chosen box and its ornaments having a total mass exceeding 360g is 0.4, determine the value of α . [5]

In another batch of wooden blocks, the manufacturer claims that the masses of the block have normal distribution with mean 130 g and standard deviation 80 g. Justify if this distribution is appropriate. [2]

9. SAJC Prelim 9758/2023/02/Q11

In this question you should state the parameters of any normal distributions you use.

On Kaypoh Island, the heights of men and women, in cm, are modelled as having independent normal distributions with means and standard deviations as shown in the table below.

	Mean (cm)	Standard deviation (cm)
Men	173	10
Women	165	σ

- (i) Find the probability that the height of a randomly chosen man is within 4 cm of the mean. [2]
- (ii) Two women are chosen at random. Suppose the probability that one of them has a height less than 165 cm and the other has a height more than 160 cm is 0.7, calculate the value of σ . [3]

It is given instead that $\sigma = 9$ for the rest of the question.

- (iii) Find the probability that a randomly chosen woman is taller than a randomly chosen man. [2]
- (iv) Do you think that the probability calculated in (iii) is equivalent to the probability that a married woman is taller than her husband? Explain your answer clearly. [1]
- (v) Find the probability that the total height of 3 randomly chosen women exceeds twice the height of a randomly chosen man by at most 1 metre. [4]

10. TJC Prelim 9758/2023/02/Q9

A stall owner sells 2 kinds of poultry namely chicken and turkey.

The masses, in kg, of chickens and turkeys are modelled as having independent normal distributions with means and standard deviations as shown in the table.

	Mean mass	Standard deviation
Chickens	2.2	0.5
Turkeys	10.5	2.1

Chickens and turkeys with masses exceeding 3.5 kg and 14.5 kg respectively are considered as too big. Chickens and turkeys with masses smaller than 1 kg and 5 kg respectively are considered as too small.

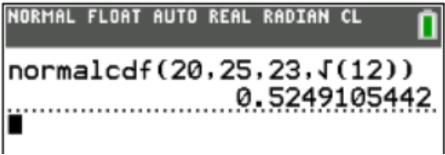
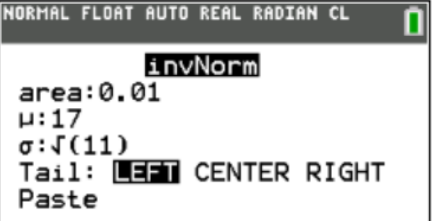
- Find the probability that a randomly chosen chicken is neither too small nor too big. [1]
- Given that a randomly chosen chicken has a mass between 3.2 kg and 3.7 kg, find the probability that it is too big. [3]
- Two chickens and one turkey are chosen randomly. Find the probability that at most one of them is too big, giving your answer correct to 5 significant figures. [4]
- Find the probability that the difference in masses between a turkey and thrice the mass of a chicken is more than 0.3kg. [4]

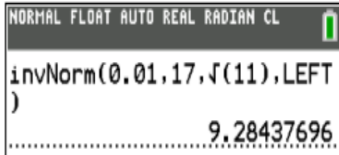
Answers

- ~~(aii) 0.748~~ Q4 (b) $\sigma = 8.01$; $\mu = 44.3$ (ci) 515 (cii) $m = 22$
- (i) 90 (iii) 0.593 (iv) 0.0401 (vi) 0.108
- (i) $\sigma = 2.80$ (ii) 1
- Q1 (i) 0.345 (ii) 0.612 (iii) 12.3, 0.475 (iv) 55.7
- (a) 1.025 (b) 0.996 (c) 2.16 (d) 0.347
- (ii) 0.605 (iii) 0.478 (iv) 0.773 (v) 0.126 (vi) 137 (vii) 0.163
- (ii) $a < 23.7$ (iii) 0.00102 (iv) 0.234
- (ii) 0.252
- (i) 0.311 (ii) $\sigma \approx 9.53$ (iii) 0.276 (v) 0.0267
- (a) 0.987 (b) 0.155 (c) 0.99971 (d) 0.970

Skill Set

Chapter S3

No.	Skills	Examples of questions involving the skills
1.	Using normalcdf in GC to evaluate probabilities when given questions involving normal distributions.	(Lecture Notes Example 2i) Given that $X \sim N(23, 12)$, state the expectation and the standard deviation of X. Evaluate $P(20 \leq X \leq 25)$. Solution Compare $X \sim N(23, 12)$ with $X \sim N(\mu, \sigma^2) \Rightarrow E(X) = \mu = 23$ and $\sigma^2 = 12 \Rightarrow \sigma = \sqrt{12}$. Using TI84+: $P(20 \leq X \leq 25) = 0.525$ 
2.	Using InvNorm in GC to find unknown in the value of the random variable.	(Lecture Notes Example 4i) Given $X \sim N(17, 11)$, find the values of a and b such that $P(X < a) = 0.01$ Solution Given $P(X < a) = 0.01$, from GC, $a = 9.28$. ① Go to the home screen. Press 2nd [DISTR], select 3:invNorm(, press ENTER. ② Key in 0.01 as the area (which is the probability required), 17 as the mean and $\sqrt{11}$ as the standard deviation. 

No.	Skills	Examples of questions involving the skills
		③ Press ☐ to move the cursor to Paste and press ENTER to paste the command onto the home screen. And press ENTER again to obtain the answer for a. 
3.	Use standardizing to find unknowns when they are in the parameters. (Unknown μ or σ)	(Lecture Notes Example 8) An orchard produces mangoes which have masses (in grams) that follow a normal distribution $N(190, \sigma^2)$. Thirty percent of the mangoes produced are heavier than 220 g. Find the value of σ. Solution Let X be the mass of mangoes produced by the orchard. Then $X \sim N(190, \sigma^2)$. It is given that $P(X > 220) = 0.3$. $P\left(\frac{X - \mu}{\sigma} > \frac{220 - \mu}{\sigma}\right) = 0.3$ (subtract $\mu = 190$ and divide by σ for each term) $P\left(Z > \frac{220 - 190}{\sigma}\right) = 0.3$ (replace $\frac{X - \mu}{\sigma}$ by Z) $P\left(Z > \frac{30}{\sigma}\right) = 0.3$ $1 - P\left(Z \leq \frac{30}{\sigma}\right) = 0.3$ $P\left(Z \leq \frac{30}{\sigma}\right) = 0.7$ $\frac{30}{\sigma} = 0.5244005101 \Rightarrow \sigma = 57.2$
4	Use properties of expectation and variance to calculate the distribution of linear combination of Normal Random Variables.	(i) $E(aX + b) = a E(X) + b$, (ii) $E(aX \pm bY) = a E(X) \pm b E(Y)$, (iii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$, (iv) $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
5	Using modulus to solve questions involving “Difference” or “within”	(Lecture Notes Example 11iii) For a particular subject, the examination marks of the boys are normally distributed with a mean of 55 and standard deviation 11. The marks of the girls are also normally distributed with a mean of 58 and standard deviation 8. If one boy and one girl are chosen at random from the

No.	Skills	Examples of questions involving the skills
		complete list of candidates, calculate the probability that the difference between the two marks is more than 20 Solution Let X be the examination mark of a boy and Y be the examination mark of a girl. Then $X \sim N(55, 11^2)$ and $Y \sim N(58, 8^2)$. The difference between the two marks is expressed as $Y - X$. $E(Y - X) = E(Y) - E(X) = 3$ and $\text{Var}(Y - X) = \text{Var}(Y) + (-1)^2 \text{Var}(X) = 185$ Then $Y - X \sim N(3, 185)$. $P(Y - X > 20) = P(Y - X > 20) + P(Y - X < -20) = 0.151$
6	Understand the difference between in n times a random variable and sum of n independent random variables.	(Lecture Notes Example 14i-ii) The mass (in kg) of a salmon fish is a random variable with the distribution $N(2, 0.25)$. The mass (in kg) of a tuna fish is a random variable with the distribution $N(3, 0.64)$. (i) Find the probability that the mass of 4 salmon fish is more than 3 times the mass of a tuna fish. (ii) Find the probability that the mass of 4 salmon fishes is more than that of 3 randomly chosen of tuna fishes. Solution Let S be the mass of a salmon fish and T be the mass of a tuna fish. Then $S \sim N(2, 0.25)$ and $T \sim N(3, 0.64)$. (i) $E(S_1 + S_2 + S_3 + S_4 - 3T) = 4(2) - 3(3) = -1$ and $\text{Var}(S_1 + S_2 + S_3 + S_4 - 3T) = 4(0.25) + (-3)^2(0.64) = 6.76$ Then $S_1 + S_2 + S_3 + S_4 - 3T \sim N(-1, 6.76)$. $P(S_1 + S_2 + S_3 + S_4 - 3T > 0) = 0.350$ (ii) $E(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3)) = 4(2) - 3(3) = -1$ and $\text{Var}(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3))$ $= \text{Var}(S_1 + S_2 + S_3 + S_4) + (-1)^2 \text{Var}(T_1 + T_2 + T_3)$ $= 4(0.25) + 3(0.64) = 2.92$ Then $S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3) \sim N(-1, 2.92)$. $P(S_1 + S_2 + S_3 + S_4 - (T_1 + T_2 + T_3) > 0) = 0.279$.

7.	Understand the difference between dependent variables and independent variables.	(Tutorials Q1ii vs 6vi) 1(ii). In order to protect them from rusting, the spheres are given a coating which increases the mass of each sphere by 10%. (ii) Find the probability that the mass of a coated sphere is between 21.5 and 22.45 grams. State the distribution you use and its parameters. $X = 1.1S \sim N(1.1 \times 20, 1.1^2 \times 0.5^2)$ 6(vi). In order to protect the bulbs in transit each bulb is surrounded by padding before being packed in a box. The mass of the padding is modelled as 30% of the mass of the bulb. (vi) The probability that the total mass of a box containing a bulb and padding is more than k grams is 0.9. Find k. Let A be the mass of the padding. $A = 0.3X \sim N(0.3 \times 50, 0.3^2 \times 1.5^2)$ i.e. $A \sim N(15, 0.45^2)$ Let $T = A + X + Y \sim N(15 + 50 + 75, 0.45^2 + 1.5^2 + 2^2)$ i.e. $T \sim N(140, 6.4525)$ Note that A and X are independent, i.e. in each box, it is $0.3X_1 + X_2 + Y$ and not $1.3X + Y$.
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Normal Distribution: Supplementary Exercise

1. [2016/HCI/MYE/P2/3]

The random variable X is normally distributed with mean μ and standard deviation σ .

- (i) Find the probability that X differs from the mean by less than 2σ . [3]
- (ii) Given that $P(X < 8) = 0.25$ and $P(X \geq 14) = 0.1$, find the values of μ and σ . [3]

2. 2016/RI/MYE/P1/9

In this question, you should state clearly the values of the parameters of any normal distribution you use.

At a car service centre, two service packages are offered: the Essential package and the Premium package. The time taken (in minutes) to service a car under these packages are independent and normally distributed, with means and standard deviations as shown in the table below:

	Mean	Standard Deviation
Essential package	110	15
Premium package	160	25

- (i) Find the probability that the difference in time taken to service a randomly chosen car under the Premium package and a randomly chosen car under the Essential package is less than 10 minutes. [3]
- (ii) Find the probability that the time taken to service a randomly chosen car under the Premium package is more than twice the time taken to service a randomly chosen car under the Essential package. [3]
- (iii) Determine the service time t such that only 10% of cars serviced under the Premium package will take more than t minutes. [1]
- (iv) After new work processes were put in place, the time taken to service a car under the Essential package is reduced by 5% while the time taken to service a car under the Premium package is reduced by 10%.

Find the probability that a total of at most 6 hours is needed to service two randomly chosen cars under the Essential package and a randomly chosen car under the Premium package. [4]

3. SAJC Prelim 9758/2017/02/Q10

A factory manufactures round tables in two sizes: small and large. The radius of a small table, measured in cm, has distribution $N(30, 2^2)$ and the radius of a large table, measured in cm, has distribution $N(50, 5^2)$.

- (i) Find the probability that the sum of the radius of 5 randomly chosen small tables is less than 160 cm. [2]
- (ii) Find the probability that the sum of the radius of 3 randomly chosen small tables is less than twice the radius of a randomly chosen large table. [2]
- (iii) State an assumption needed in your calculation in part (ii). [1]

A shipment of 12 large tables is to be exported. Before shipping, a check is done and the shipment will be rejected if there are at least two tables whose radius is less than 40 cm.

- (iv) Find the probability that the shipment is rejected. [3]

The factory decides now to manufacture medium sized tables. The radius of a medium sized table, measured in cm, has distribution $N(\mu, \sigma^2)$. It is known that 20% of the medium sized tables have radius greater than 44 cm and 30% have radius of less than 40 cm.

- (v) Find the values of μ and σ . [4]

4. TPJC Prelim 9758/2017/02/Q8

A manufacturing plant processes raw material for a supplier. An order placed with the plant is considered to be a bulk order when a worker is expected to process more than 300 kg (kilograms) of raw material.

Albert uses a machine to process X kg of raw material and Bob uses a separate machine to process Y kg of raw material on a working day. X and Y are independent random variables with the distributions $N(296, 8^2)$ and $N(290, 12^2)$ respectively.

- (i) Find the probability that Albert processes more than 300 kg of raw material on a randomly selected working day. [2]
- (ii) Find the probability that, over a period of 15 independent working days, there are exactly four working days on which Albert processes more than 300 kg of raw material. [2]
- (iii) Find the probability that the total amount of raw material Bob processes over two working days exceeds twice the amount of raw material Albert processes on one working day. [4]

The plant receives a bulk order and Albert wants to have a probability of at least 0.95 of meeting the order.

- (iv) This can be done by changing the value of μ , the mean amount of raw material Albert processes using the machine, but the standard deviation remains unchanged. Find the least value of μ . [3]

5. MI 2018/PU3/P2/11

Belgian chocolate is renowned across the world for its quality. At a Great Singapore Sale food fair at Takashimaya Shopping Centre, several brands are available.

The masses, in grams (g), of packets of Godiva chocolate and Guylian chocolate are modelled as having independent normal distributions. The means and standard deviations of these distributions, and the selling prices, in cents per gram, are shown in the table.

	Mean (g)	Standard Deviation (g)	Selling price (dollars per g)
A packet of Godiva	184.2	5.2	0.03
A packet of Guylian	80	3.1	0.02

- (i) State, with a reason, whether a normal model is likely to be appropriate for the mass of a packet of chocolate chosen at random from all the packets available. [1]
- (ii) Find the probability that both a randomly chosen packet of Godiva has a selling price exceeding \$5.50 and a randomly chosen packet of Guylian has a selling price exceeding \$1.50. [3]
- (iii) Find the probability that the total selling price of a randomly chosen packet of Godiva and a randomly chosen packet of Guylian is more than \$7.00. [3]
- (iv) Explain why the answer to part (iii) is greater than the answer to part (ii). [1]
- (v) Packets of Neuhaus chocolate have mass with mean 75 g and standard deviation 2.1 g. A random sample of 50 packets of Guylian chocolate and a random sample of 50 packets of Neuhaus chocolate are taken. Find the probability that the difference in the sample means is less than 4 g. this requires Central Limit theorem, S4 [3]

To boost sales, Takashimaya organises a lucky draw with a fully-paid holiday to Belgium for two as its top prize. Only customers who purchase a packet of Godiva chocolate qualify for the lucky draw, and the number of lucky draw coupons received depends on the mass of the packet. If the mass of the packet is less than 186 g, the customer receives one coupon. If the mass of the packet is at least 186 g but less than 192 g, the customer receives two coupons. Otherwise, the customer receives m coupons.

The number of lucky draw coupons a customer receives for a packet of Godiva chocolate is denoted by L .

- (vi) Find, in terms of m , the probability distribution of L . [2]
- (vii) Given that the expected number of lucky draw coupons a customer receives for a packet of Godiva chocolate is 1.90, find m to the nearest integer. Hence find $\text{Var}(L)$. [4]

6. EJC Prelim 9758/2018/02/Q10

In this question, you should state clearly the distribution of any random variables that you define. The volume, S , in ml, of perfume in a randomly chosen small bottle has mean 20 and variance σ^2 .

- (i) If $\sigma = 15$, explain why S may not be appropriately modelled by a normal distribution. [2]

It is now assumed that S follows a normal distribution.

- (ii) Given that 6.68% of the small bottles contains more than 23 ml of perfume, find the value of σ . [2]

For the rest of the question, the volume of perfume, S , in ml, in a randomly chosen small bottle follows the distribution $N(20, 4)$ and the volume of perfume, L , in ml, in a randomly chosen large bottle follows the distribution $N(100, 25)$.

- (iii) Calculate the probability that 6 randomly chosen small bottles and 9 randomly chosen large bottles contain a total volume of at least 1 litre of perfume. [3]
- (iv) Calculate the probability that the volume of perfume in a randomly chosen large bottle differs from 6 times the volume of perfume in a randomly chosen small bottle by more than 25ml. [3]
- (v) State, in this context, an assumption needed for your calculations in parts (iii) and (iv). [1]

7. TPJC Prelim 9758/2018/02/Q10

- (a) A and B are independent random variables with the distributions $N(25, 20)$ and $N(\mu, \sigma^2)$ respectively. It is known that $P(B < 12) = P(B > 19)$ and $P(A > B) = 0.68$. State the value of μ and calculate the value of σ . [4]
- (b) There are two vets Jerry and Mary who attend to customers with sick pets in a small vet clinic. The time taken for Jerry to attend to a customer follows a normal distribution with mean 10.1 minutes and standard deviation 0.8 minutes and the time taken for Mary to attend to a customer follows an independent normal distribution with mean 10.3 minutes and standard deviation 0.75 minutes.
- (i) Find the probability that among three randomly chosen customers attended to by Jerry, one took less than 10 minutes while the other two each took more than 10.5 minutes. [2]
- (ii) Find the probability that the total time Jerry took to attend to two randomly chosen customers is less than twice the time Mary took to attend to one randomly chosen customer by at least 3 minutes. State the distribution you use and its parameters. [3]
- (iii) Jerry and Mary each attended to two randomly chosen customers. Find the probability that the difference in the total time taken by Jerry and Mary to attend to their two customers is more than 2 minutes. State the parameters of any distribution you use. [3]

8. NJC 2019/BT2/II/8

A factory supplies roasted coffee powder sold in packets. The mass of roasted coffee powder in a randomly chosen packet is taken to have a normal distribution. It is found that 5.5% of the packets contain less than 1450 g roasted coffee powder and 11.5% of the packets contain more than 1520 g of roasted coffee powder.

- (i) Show that the mean of the distribution, to nearest integer, is 1490 g. Find the standard deviation. [3]

Only 3% of the packets contain more than m g of roasted coffee powder.

- (ii) Find the probability that out of three randomly chosen packets of roasted coffee powder, two packets each contains less than the mean mass of the roasted coffee powder and the remaining packet contains between the mean mass and m g of the roasted coffee powder. [3]

A wholesaler wants to purchase 500 packets of roasted coffee powder from the factory. A random sample of 20 packets of roasted coffee powder is taken from the batch of 500

packets. If the sample contains more than 4 packets of coffee powder each with a mass less than 1450 g, the wholesaler will reject the batch of 500 packets.

- (iii) State, in context, two assumptions needed for the number of packets of roasted coffee powder each with a mass less than 1450 g to be well modelled by a binomial distribution. [2]

Assume the assumptions made in part (iii) hold.

- (iv) Calculate the probability that the wholesaler accepts the batch of 500 packets given that there is at least one packet in the sample with a mass less than 1450 g. [4]

9. ACJC JC2 Prelim 9758/2019/02/Q10

In this question you should state the parameters of any distributions you use.

Mary runs a noodle stall by herself. The noodles are prepared to order, so she prepares the noodles only after each order, and will only take the next customer's order after the previous customer is served his noodles.

The time taken for a customer to place an order at the stall follows a normal distribution with mean 60 s and standard deviation σ s. The time taken for Mary to prepare and serve a customer's order also follows a normal distribution, with mean 300 s and standard deviation 50 s.

The probability that a customer takes not more than 40 s to place an order is 0.16. Show that $\sigma = 20.11$, correct to 2 decimal places. [2]

- (i) Let A be the probability that a randomly chosen customer takes between 57 s and 63 s to place an order with Mary, and B be the probability that a randomly chosen customer takes between 49 s and 55 s to place an order with Mary. Without calculating A and B , explain, with the aid of a diagram, how A and B compare with each other. [2]
- (ii) There is a 0.1% chance that a randomly chosen customer has to wait for more than k seconds for his noodles to be served after placing his order. Find k . [1]
- (iii) A man visits the noodle stall on 10 separate occasions. Find the probability that his average waiting time per visit after placing his order is less than 4.5 minutes. [2]
- (iv) In an effort to shorten wait time, Mary improves the ordering and cooking processes such that the time taken for a customer to place an order is reduced by 5%, while the time taken to prepare and serve a customer his noodles is reduced by 10%. Find the largest number of customers she can serve in 1 hour for at least 80% of the time. State one assumption you made in your calculations. [5]

10. RVHS JC2 Prelim 9758/2019/02/Q12

An orchard produces apples and pears.

- (i) The masses of apples produced by the orchard have mean 70 grams and standard deviation 40 grams. Explain why the masses of apples in the orchard are unlikely to be normally distributed. [1]
- (ii) The masses of pears produced by the orchard are normally distributed. It is known that one-third of the pears produced weigh less than 148 grams, and one-third weigh more than 230 grams. Find the mean and standard deviation of the masses of pears produced in the orchard. [4]

During each production period, the orchard produces 300 apples and 400 pears and the owner is able to sell all the apples at \$0.005 per gram and all the pears at \$0.008 per gram. It is further known that the owner needs to spend a fixed amount of \$ c as the operating cost for each production period and that the orchard makes a profit 90% of the time.

- (iii) Explain why the total mass of the 300 apples is approximately normally distributed. [1]
this requires Central Limit Theorem, S4
- (iv) By letting R be the amount of money collected from selling the 300 apples and 400 pears, find the distribution of R and hence, determine the value of c . [6]

11. TJC Prelim 9758/2020/02/Q8

In a TV sports entertainment competition, Spartan Warrior, each contestant has to complete two obstacle stations X and Y in the shortest time possible. It has been found that the time taken to complete station X is normally distributed with mean μ minutes and variance 1.29 minutes², and the time taken to complete station Y is also normally distributed with mean 6.5 minutes and variance 1.06 minutes². Assume that each contestant begins station Y immediately after completing station X .

- (i) If the probability that a randomly chosen contestant takes more than 22.5 minutes to complete both stations is less than 0.0102, find the range of values that μ can take. State an assumption you made in your calculations. [4]

For the remainder of the question, assume that $\mu = 12.3$.

- (ii) Find the probability that the time taken by a randomly chosen contestant to complete station X is greater than twice the time taken to complete station Y . [3]

Assume that a contestant begins immediately after the previous contestant completes both stations.

- (iii) Find the maximum number of contestants that can be invited to a recording session if the probability that the total time taken for all contestants to complete both stations is within 7 hours is more than 0.99. [3]
- (iv) During the broadcast of each episode, the sports channel will broadcast 3 randomly chosen contestants completing both stations without editing. Find the probability that, in one episode, the time taken by one of the 3 randomly chosen contestants to complete both stations is less than 17 minutes and the time taken by the remaining 2 contestants to complete both stations is more than 20 minutes each. [2]

12. NYJC Prelim 9758/2020/02/Q10

In this question you should state clearly all the distributions that you use, together with the values of the appropriate parameters.

The Nanyang Symphony Orchestra plays a mixture of classical and jazz music pieces. The length, in minutes, of each piece of music are modelled as having independent normal distributions. The means and standard deviations of these distributions are shown in the following table.

	Mean	Standard deviation
Classical piece	12	2.3
Jazz piece	7	1.5

The orchestra organises a challenge whereby it will be playing a repertoire of 5 classical and 4 jazz pieces which are randomly chosen by a live audience.

- (i) Calculate the probability that the total length of playing the 5 randomly chosen classical pieces is less than 50 minutes. [2]
- (ii) Find the probability that the mean length of these 9 randomly chosen pieces in the repertoire is less than 8 minutes. [2]
- (iii) The orchestra decides to have an interval of exactly one minute between the orchestra finishing one piece and starting the next piece. The challenge starts at 7.00 pm. By using your answer in part (ii), state whether it is likely that the orchestra will still be playing after 8.20 pm. Justify your answer. [2]

The orchestra plans to record the repertoire into an album.

- (iv) The orchestra estimates that the total recording time will be **twice** the total length of the 9 pieces in the repertoire and proceeds to book a slot with the recording studio. The booking contract of the recording studio states that if the time taken to record the repertoire exceeds the booked duration of x minutes, a fine will be imposed. Find the minimum duration, to the nearest minute, for which the orchestra must book the recording studio if it wants to be at least 99% sure that it will not be fined. [3]
- (v) Due to the different expertise and resources involved in recording classical and jazz pieces, the recording studio charges a rate of \$12 per minute for a classical piece and \$9.50 per minute for a jazz piece. Find the probability that the total charge of recording the 5 randomly chosen classical pieces is within \$300 of the total charge of recording the 4 randomly chosen jazz pieces. [3]

13. DHS Prelim 9758/2021/02/Q10

The queue time for customers, in minutes, at a particular hair salon has a mean of 15 minutes and a standard deviation of 10 minutes.

- (i) Explain why a normal model would not be suitable for the queue times. [1]

The time duration of a haircut for customers, in minutes, at 2 different hair salons, Qcut, an express service, and SPcut, a specialist service, are normally distributed with mean and standard deviation as given in the table.

	Mean	Standard Deviation
Qcut	9.2	σ
SPcut	20.7	3.1

- (ii) If the probability that a Qcut customer having a haircut duration of more than 10 minutes does not exceed 0.35, find the range of values σ can take up. [2]

Use $\sigma = 1.5$ for the rest of the question.

- (iii) Given that the difference between a haircut duration of a randomly chosen Qcut customer and its mean has a 95% chance of being larger than k , find the value of k . [2]
- (iv) Find the probability that the mean haircut duration of two Qcut customers and three SPcut customers is less than 15 minutes. [3]
- (v) State an assumption needed for your calculations in part (iv) to be valid. [1]

The salons fee-charging system comprises 2 components. Qcut charges a fixed component of \$3 and a variable component of \$0.50 per minute. Similarly, SPcut charges a fixed component of \$10 and a variable component of \$2 per minute.

- (vi) Find the probability that the fees paid by a randomly chosen SPcut customer is at least 6 times the fees paid by a randomly chosen Qcut customer. [3]

14. RVHS Prelim 9758/2021/02/Q10

In a certain district, the English and Maths marks for Primary 6 students in the national primary school leaving examination follow normal distributions with means and variances shown below:

Subject	Mean	Variance
English	76	25
Maths	74	σ^2

The performance in English is independent of that in Maths for Primary 6 students in the district.

- (i) Past data revealed that 1 in every 30 students in the district scored at least 85 marks in Maths in the examination. Find the value of σ correct to the nearest whole number. [2]

For the rest of the question, take σ to be the nearest whole number obtained in part (i).

- (ii) To qualify for the top secondary school in the district, a Primary 6 student will need to have the sum of his Maths mark and double of his English mark to be at least 250. Given that a particular student qualifies for the top secondary school, find the probability that he scored more than 85 marks in each of the English and Maths examination. [4]
- (iii) The Education Board of the district decides to study if there is any significant difference in the Primary 6 students' performance for the 2 subjects in the examination. A sample of 5 students is chosen from School A in the district and the sum X of their English marks is obtained. Another sample of 5 students is chosen from School B in the same district and the sum Y of their Maths marks is obtained. Find the probability that the difference of X and Y is more than 30 marks. [4]

A student scoring more than 80 marks in English qualifies for an enrichment course. A sample of n students are chosen. Find the least value of n such that the probability of more than 4 students qualifying for the enrichment course is more than 0.15. [4]

15. TJC Prelim 9758/2022/02/Q10

A workshop employs craftspeople to make wooden souvenirs for sale at a tourist attraction. Typically, the time taken to make a wooden souvenir by a skilled craftsman follows a normal distribution with a mean of 1.2 hours and a standard deviation of 10 minutes. A workday at the workshop is 8 hours long (with a lunch break given separately). It may be assumed that a skilled craftsman will start making the next souvenir once the previous one is completed.

- (i) Find the probability that a skilled craftsman is able to make 7 wooden souvenirs in a workday. [2]
- (ii) Two skilled craftspeople start work at the same time. Find the probability that they make their first wooden souvenir of the workday within 15 minutes of each other. [3]
- (iii) Find the probability that a skilled craftsman is unable to make the 7th souvenir in a workday. Hence find the probability that a skilled craftsman makes exactly 6 wooden souvenirs in a workday. [3]
- (iv) The time taken by an apprentice of a particular skilled craftsman to make a wooden souvenir is 1.5 times of the time taken by the skilled craftsman. Find the probability that the total time taken to make 3 wooden souvenirs by the apprentice is longer than 4 times the time taken by the skilled craftsman to make a wooden souvenir.
State the assumption needed for your working. [4]